Session Objectives

1. Define nosocomial infections and their prevalence.
2. Recognize the most common nosocomial etiological agents with emphasis on nosocomial pneumonia.
3. Explain the importance of infection control and major causes of nosocomial infections.
4. Describe what infection control programs do, including the importance of handwashing, isolation and precautions, surveillance/reporting and cluster/outbreak investigation.
5. Appreciate that choosing the appropriate isolation/precautions requires in-depth understanding of microbial transmission.
6. Discuss the challenges nosocomial diseases present for patient care.

Source: Course Syllabus

How to Prevent NIs: Infection Control!

- Reduce spread of infection from:
  - Patient to patient
  - Patient to worker
  - Worker to patient
  - Worker to worker
  - Visitor to patient to worker, etc.
- Identify (& report) nosocomial infections
  - Surveillance in (& interventions to reduce):
    - High risk areas (ICUs)
    - High risk infections (CLABSI, CAUTI, CDI, HCAP, VAP, SSI)
- Employee health issues (vaccination)
- ***Expanded Issues:*** *Reportable Diseases, Emerging pathogens/ bioterrorism response (H1N1), Antibiotic stewardship*

Birth of Infection Control

- Ignaz Semmelweis (1849)
  - In postpartum women with childbed fever (puerperal fever due to group A *Streptococci*) the mortality was higher when delivered by obstetricians & students and lower when delivered by midwives
  - Obstetricians & students frequently performed deliveries **after autopsies** (often on patients who died of childbed fever & other infectious diseases)
  - Instituted **hand washing** using a chlorine solution, **decreasing postpartum mortality from 11% to 1%**



Handwashing

- Single most effective intervention to reduce the cross transmission of nosocomial infections
- Must be "bacteriologically effective"
- Wash hands before any procedure in which gloves and forceps are necessary
- After contact with infected patient or one colonized with multidrug-resistant bacteria
- After touching infected material
- Use soap & water (preferably disinfectant soap)
- More prolonged & thorough scrub before surgery

Isolation/Precautions

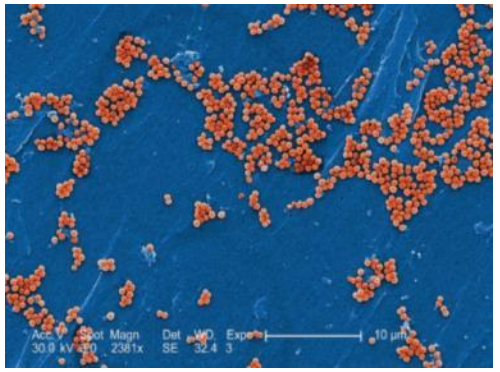
- Type 1: Prevent transmission of pathogen from patient to healthcare worker, e.g., influenza
  - Prevents
    - Healthcare worker acquisition
    - Transmission from patient to patient
- Type 2: Prevents transmission from healthcare worker to patient at risk, e.g., NICU, bone marrow transplant units
  - aka reverse precautions or protective isolation

Isolation/Precautions

- Standard precautions
  - Minimum set of infection prevention measures performed with all hospitalized patients
- Transmission-based precautions
  - Supplement standard precautions for patients known infected or colonized with certain transmissible pathogens

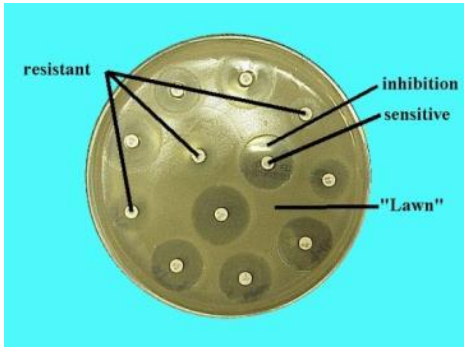
Staphylococci

- Gram +ve cocci
- Normal flora of humans:
  - Nasal membranes/skin
- Opportunistic infections:
  - Focal damage
    - 53% purulent wounds
    - 61% abscesses
  - Systemic disease
- Nosocomial infections
- MRSA is a strain resistant to the beta lactam antibiotic methicillin



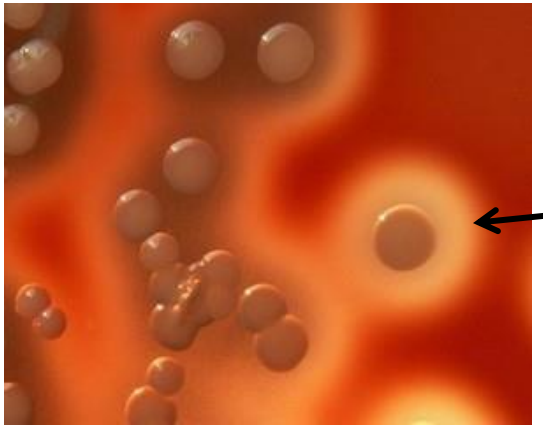






Characteristic of the genus *Staphylococcus*

- All members are highly resistant to drying, extremes in pH, & high temperatures (Biofilm formation)
- Increase in community acquired cases, with infections associated with schools, prisons, & shared athletic equipment
- MRSA was responsible for 94,000 life-threatening infections & nearly 19,000 deaths in the US
- *S. epidermidis* is a common cause of nosocomial infections in immunocompromised patients



S. aureus is β-hemolytic on blood agar

TABLE 32.1 Differentiation of staphylococcal species

	S. aureus	*S. epidermidis*
Growth on media containing 7.5% NaCl	+	+
Beta hemolysis	+	–
Mannitol fermentation	+	–
Coagulation of plasma	+	–
Reaction to novobiocin	Sensitive	Sensitive

Enterococcus spp. (Group D *Streptococci*)

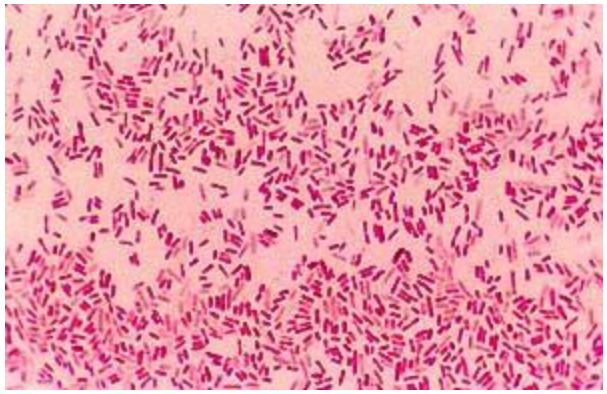
- Lancefield Group D (+) cell wall Ags (teichoic acid)
- Normal colonists of GI
- Transmitted person-to-person (direct contact)
- Occasionally on medical devises (fomites)
- Cause opportunistic urinary, wound, & skin infections, particularly in debilitated persons
- Cephalosporin and other antibiotics select for the strains
- Vancomycin-resistant *Enterococci* (VRE) are an important threat in nosocomial infections
  - 30% of *Enterococci* in ICUs are VRE

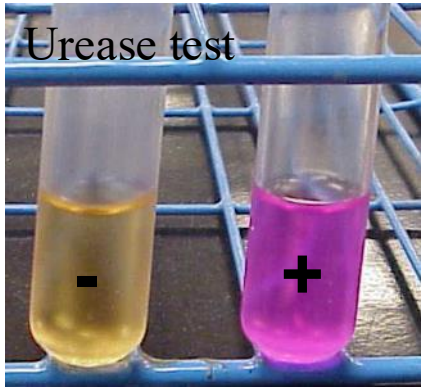
Enterococcus spp.

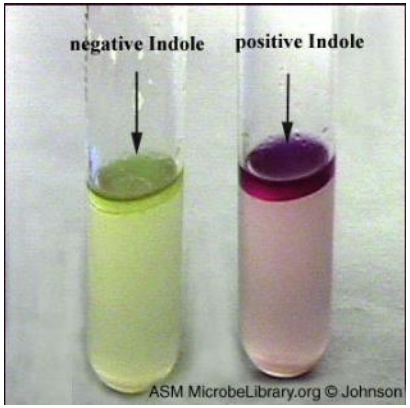
- *E. faecalis* is most common pathogen (85-90% of cases) but in HAI *E. faecium* is more common
- *E. faecium* more antimicrobial resistant than *E. faecalis* especially to vancomycin (multiple phenotypes)
- Intrinsic resistance to cephalosporins, penicillins, monobactams, aminoglycosides & fluoroquinolones
- Inhibited by ampicillin but not killed

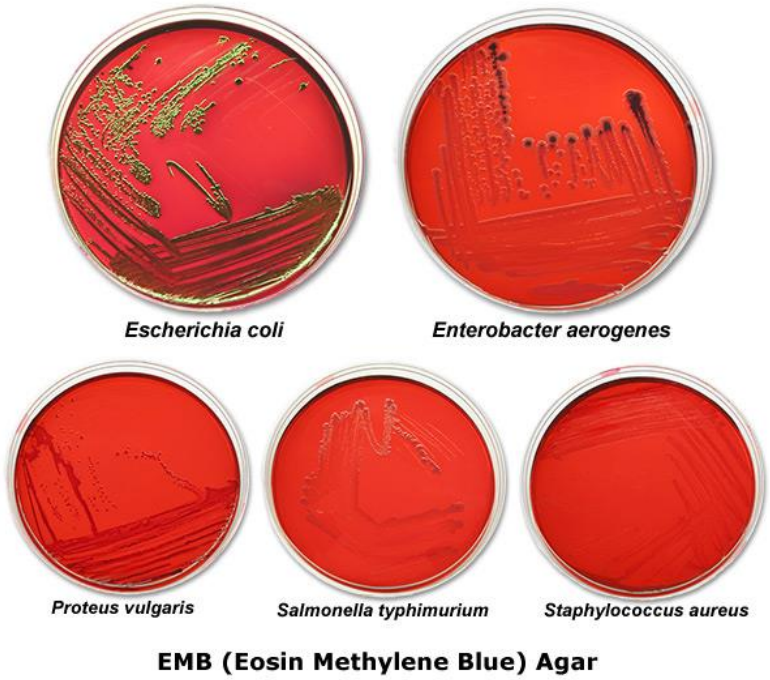
Escherichia coli

- Gram -ve rod
- UTIs, bacteremia, pneumonia, meningitis, & wound infections
- Ferments lactose (on EMB, black colonies and metallic green sheen or MAC agar)
- Indole +ve and urease -ve
- *E. coli* is the most abundant facultative anaerobe in the colon & feces
- 3 antigens (Ag) are used to type *E. coli*: O-Ag (LPS), H-Ag (flagella), and K (capsular). e.g. ***E. coli* O157H7**



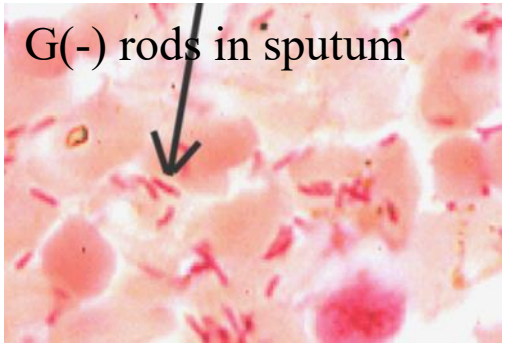


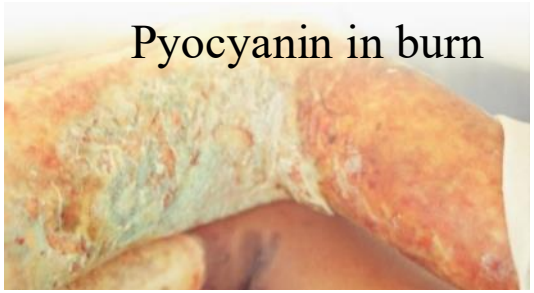




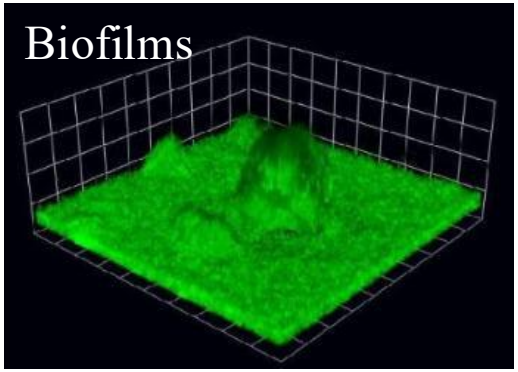
Pseudomonas species

- Can involve almost any part of body
- Especially a pathogen in immunosuppressed hosts- malignancy, diabetes, chemotherapy, other immunosuppression
  - Most common pathogen isolated from patients who have been hospitalized longer than 1 week
  - A frequent cause of nosocomial infections such as pneumonia, urinary tract infections (UTIs), & bacteremia
  - Chronic infection of the lower respiratory tract prevalent among patients with CF
  - Burn victims









Klebsiella species: characteristics

- Genus *Klebsiella* (*Enterobacteriaceae*)
- Gram -ve
- Prominent polysaccharide capsule (accounts for large appearance on Gram stain & mucoid colonies)
- Lactose-fermentor, urease +ve, & indole -ve
- 2 types of Ags on cell surface:
    - LPS (O antigen) - 9 O antigens
    - Capsular polysaccharide (K antigen) - 77 K antigens
- 3 main species (7 similar DNA homology)
    - *K. pneumoniae* - most medically important
    - *K. oxytoca & K. rhinsclermatis*

Klebsiella: HAIs

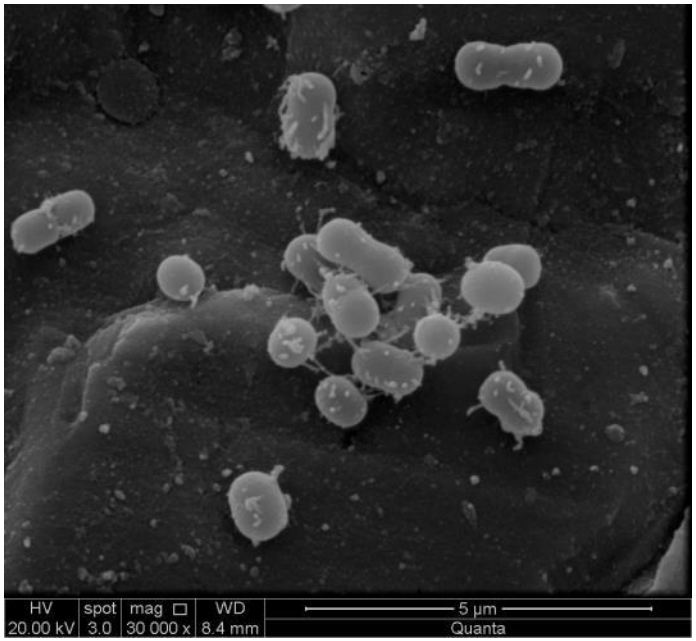
- Syndromes*:

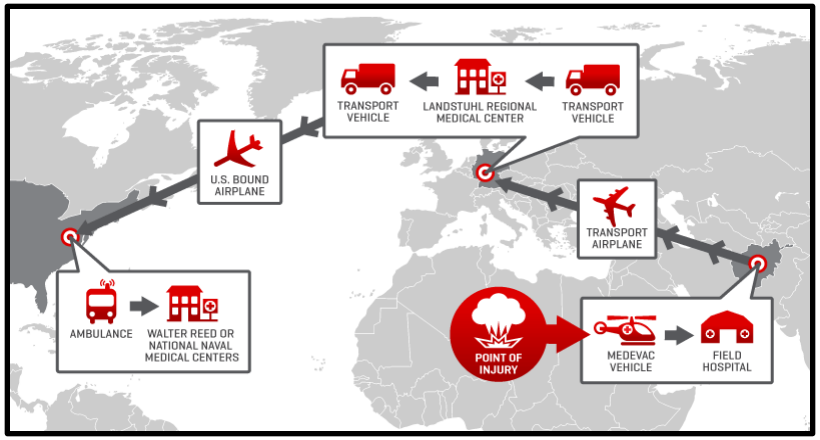
Pneumonias	Bacteremia
Thrombophlebitis	Urinary tract infection (UTI)
Cholecystitis	Diarrhea
Upper respiratory infection	Wound infection
Osteomyelitis	Meningitis

  - *Sepsis and shock may follow with entry of organism into blood from a focal source
- Increasing likelihood HAI with *Klebsiella* spp.:
  - Invasive devices
  - Contaminated respiratory support equipment
  - Use of urinary catheters
  - Use of antibiotics
- *K. oxytoca* implicated in neonatal bacteremia-premature infants and in neonatal ICUs

Acinetobacter baumannii

- Gram -ve coccobacillus
- Generally low virulence
- Opportunistic
- Adaptable growth requirements
- Aerobic, non-motile
- Biofilm formation
- Oxidase –ve & non-fermentor
- Nasopharyngeal transient flora
- Multi-drug resistance
- Most important complication in military wound victims

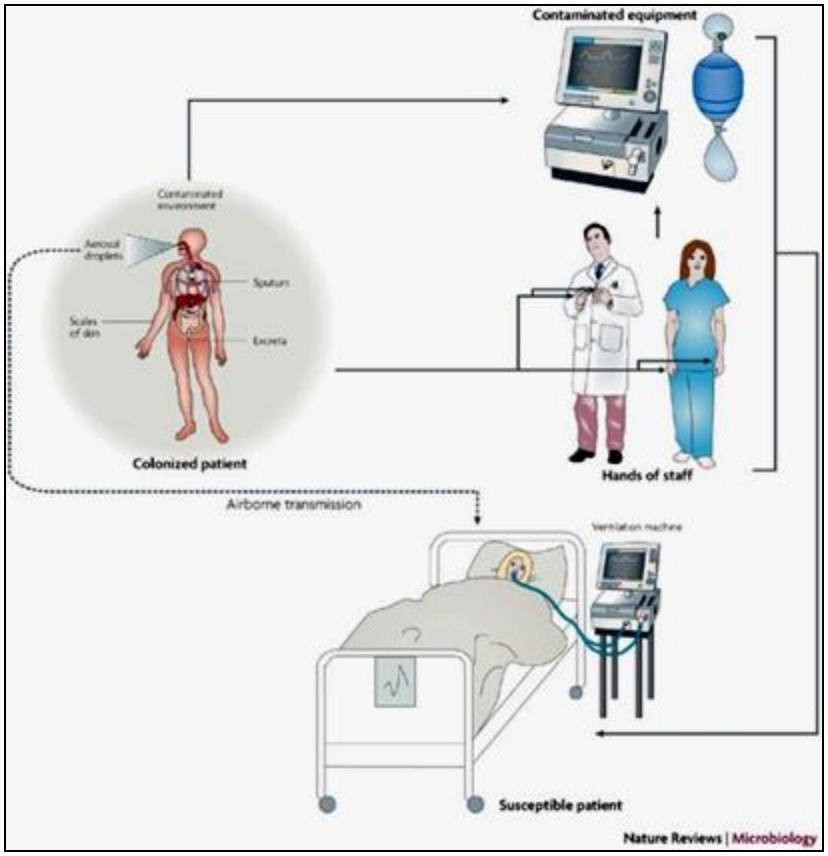




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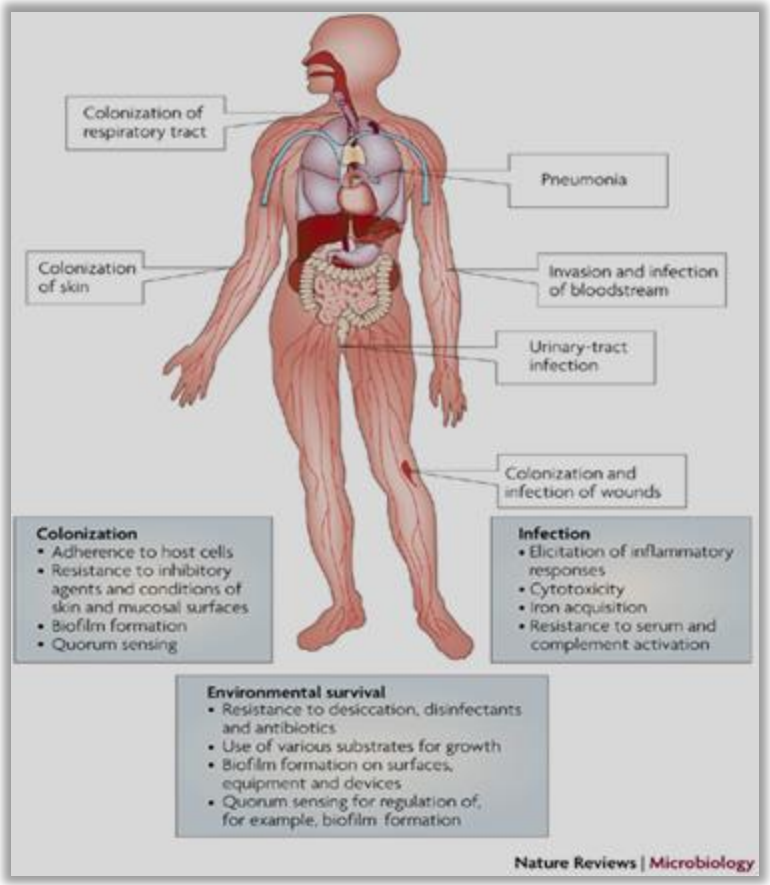
Transmission & prevalence

- Mostly found in nosocomial settings like I.V. solutions, urinary catheters, surgical drains or monitoring devices
- The bacteria can live up to 5 months on undisturbed surfaces
- Mostly enters in the body through open wounds, catheters & breathing tubes (respiratory tract)
- It affects immunocompromised individuals
- Community-acquired pneumonia affects alcoholics in tropical regions & frequently leads to systemic infection and has a mortality rate of ~50%



Colonization & Infection

- Infections usually occur in organs with high fluid content like respiratory tract, CSF, peritoneal fluid, urinary tract manifesting nosocomial pneumonia
- It can cause nosocomial meningitis, sepsis & UTI
- Also implicated in necrotizing fasciitis



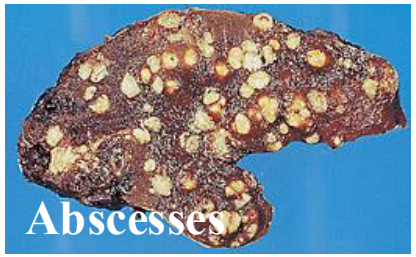


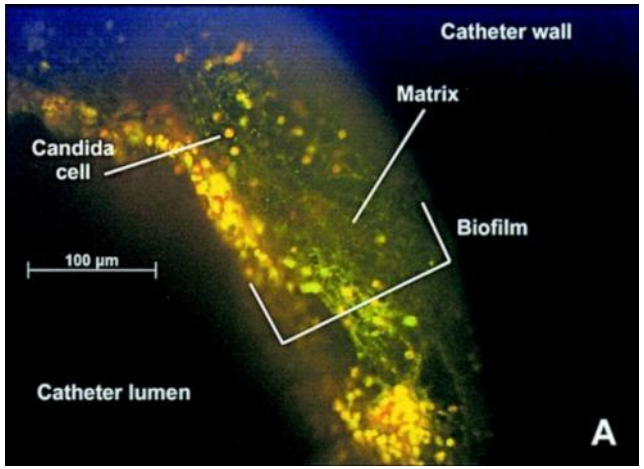


Candida species









- Most "important" fungal pathogen
- Associated with impaired immunity
- *C. albicans*: most common species (dimorphic)
- "non-*albicans*" *Candida*: increasing in incidence
- Spectrum of clinical syndromes: mild to fatal, dependent on host immune status
  - Disseminated, cutaneous and mucosal (e.g. oral thrush, vaginitis, etc.), UTIs, and neonatal candidiasis
- **Risk factors:** catheters, surgery, burns, prolonged antibiotic usage, cytotoxic therapy, parenteral hyper alimentation, pregnancy, diabetes

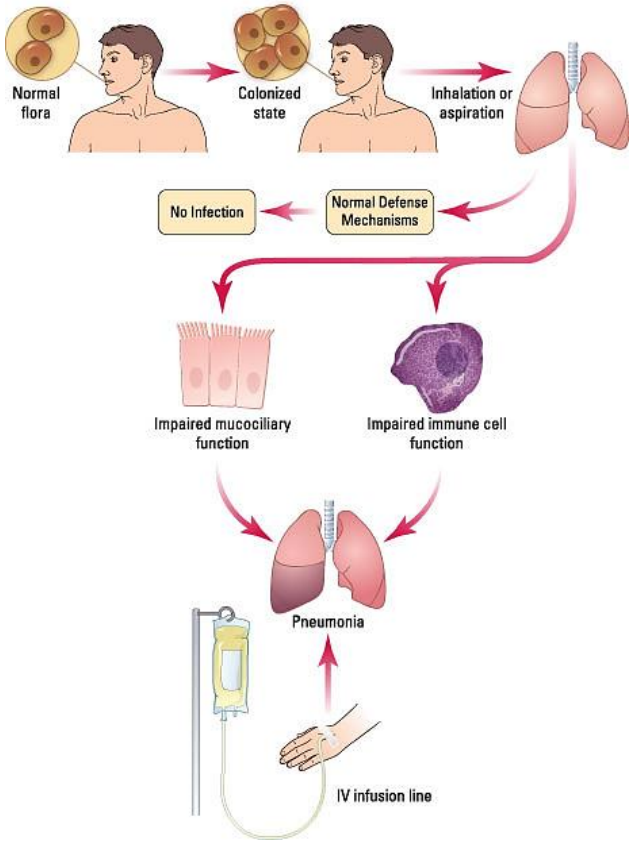
How is NP acquired?

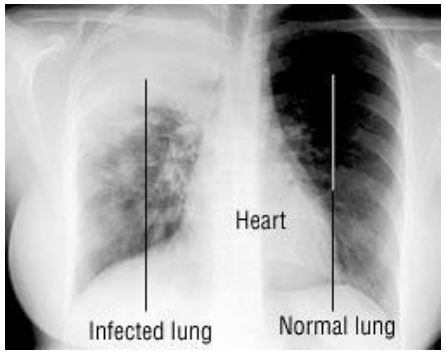
- **3 types:**

1. **Primary inhalation pneumonia** develops:
   - when microbes bypass normal respiratory defense mechanisms
   - when the patient inhales microbes that colonize the upper respiratory tract
   - when the patient is connected to respiratory support equipment (ICU)
2. **Aspiration pneumonia** occurs when the patient aspirates colonized upper respiratory tract secretions
3. **Hematogenously-acquired infections**
   - In bacterial nosocomial pneumonia, blood cultures frequently are positive if obtained early in the disease and if the patient is not already receiving antimicrobial therapy

NP

- The National Healthcare Safety Network (NHSN) reports that NP accounts for **31%** of all nosocomial infections in ICUs
- The mortality rate reaches 20–50%
- The increase in antibiotic-resistant pathogens in hospitals has contributed to the seriousness of these infections
- Respirator associated: *Klebsiella*, *Pseudomonas*, *Acinetobacter*, *Staphylococci*
- Non-respirator associated: *Legionella**





*= Can be part of the "Atypical Pneumonia" syndrome

Diagnosis, treatment & prevention

- Clinical diagnosis
  - fever, change in PO_2, change in sputum, CXR
- Microbiologic confirmation
  - Suctioned sputum sample
  - Bronchoscopy with brochoalveolar lavage
- Empiric antibiotic - clinical acumen
  - Tx based on previous cultures, usual hospital flora & susceptibilities
  - Sputum Gram stain
  - Colonization vs. infection
- Pulmonary hygiene
  - Change position every 2-hrs
    - Elevate head to 30-45 degrees
  - Deep breathing & incentive spirometry
  - Frequent suctioning
  - Bronchoscopy to remove mucous plugging

Summary

- Defined nosocomial infections and their prevalence.
- Reviewed the most common nosocomial etiological agents with emphasis on nosocomial pneumonia.
- Reviewed the importance of infection control and major causes of nosocomial infections.
- Reviewed infection control programs, including the importance of handwashing, isolation and precautions
- Reviewed appropriate isolation/precautions
- Reviewed the challenges nosocomial diseases present for patient care.